

Question

Mining minerals generates a large amount of wastewater. The steps for the highly selective iodometric determination of cobalt content are as follows:

1 Weigh 0.2517 g of arsenic trioxide standard, dissolve in 10%NaOH, and dilute to 250.0mL

2 Take 500.0mL of wastewater, evaporate and concentrate it, add a certain amount of nitric acid-hydrochloric acid, evaporate to dryness, dissolve soluble salts with an appropriate amount of water, add nitric acid and potassium chlorate again, boil, filter the precipitate, and wash the precipitate.

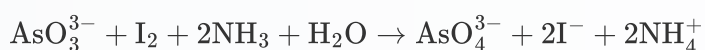
3 Concentrate the solution volume to 30 – 40mL, add 25 g of ammonium nitrate, add saturated ammonium citrate solution and an appropriate amount of ammonia water to bring the solution to pH = 9 – 9.5, transfer to 50.0mL of iodine standard solution, shake and let stand for a period of time.

4 Titrate the excess iodine with sodium arsenite solution. Add starch near the end point. Take the reading after the blue color disappears. The blank experiment consumed 30.46mL of sodium arsenite, and the titration experiment consumed 14.65mL of sodium arsenite.

Based on the above content, select the correct answer.

A. All other options are incorrect

B. The chemical equation for the titration of iodine with arsenite solution in the system is as follows:



C. It is possible to accurately weigh sublimed and purified iodine solid to prepare a standard iodine solution, and then use a $\text{Na}_2\text{S}_2\text{O}_3$ solution to replace the sodium arsenite solution for titration.

D. The concentration of Co element in the sample is 18.96mg/L

E. The concentration of Co element in the sample is 9.48mg/L

F. The concentration of Co element in the sample is 37.93mg/L

Answer

Correct Answer: **F**

Detailed Explanation

In the chemical reaction equation, the pK_a of the third ionization step of arsenic acid is 11.50, and the pK_a of the second ionization step is 6.96. The pK_a of the ammonium ion is 9.25.

CHECKPOINT

1 PTS

The pK_a of the third ionization step of arsenic acid is 11.50, and the pK_a of the second ionization step is 6.96. The pK_a of the ammonium ion is 9.25

Therefore, the existing form of arsenic acid molecules in the reaction equation should be HAsO_4^{2-} .

CHECKPOINT

1 PTS

The existing form of arsenic acid molecules should be HAsO_4^{2-}

The reaction of I_2 with $\text{S}_2\text{O}_3^{2-}$ under alkaline conditions is non-quantitative and may produce high-valent sulfur products, so it cannot be used for titration. Therefore, option C is incorrect.

CHECKPOINT

1 PTS

The reaction of I_2 with $S_2O_3^{2-}$ under alkaline conditions is non-quantitative

As(III) will be oxidized to As(V), $I_2(0)$ will be reduced to negative one valence, and Co(II) will be oxidized to Co(III). Therefore, the ratio of the reaction is as follows: $As_2O_3 \sim 2I_2 \sim 4Co^{2+}$, when the reaction ratio is judged incorrectly, option D or option E will be obtained.

CHECKPOINT

1 PTS

Reaction ratio $As_2O_3 \sim 2I_2 \sim 4Co^{2+}$

$$c(As_2O_3) = \frac{m(As_2O_3)}{V \cdot M(As_2O_3)} = 0.005089 \text{ mol} \cdot L^{-1}$$

CHECKPOINT

1 PTS

$$c(As_2O_3) = 0.005089 \text{ mol} \cdot L^{-1}$$

The difference between the blank experiment and the titration experiment consuming As(III) corresponds to the content of Co: $n(Co) = c(As_2O_3) \cdot (V_1 - V_2) \cdot 4 = 0.3218 \text{ mmol}$

CHECKPOINT

1 PTS

$$n(Co) = 0.3218 \text{ mmol}$$

$\omega(\text{Co}) = \frac{n(\text{Co}) \cdot M(\text{Co})}{V} = 37.93 \text{ mg/L}$, so the correct option is F.

CHECKPOINT

1 PTS

$$\omega(\text{Co}) = 37.93 \text{ mg/L}$$